

LaserCortex

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Chapter 1

EML Registry: Tamari Lattice Contraction

1.1 Core Types

Definition 1 (EML Tree). The type of binary trees representing configurations in the Tamari lattice. A tree is either a **Leaf** (empty configuration) or a **Node** $t_1\ t_2$ (composition of two sub-configurations).

Definition 2 (Tree Size). The number of internal nodes in a tree, used as the lattice index n in T_n .

Definition 3 (Single-Step Contraction). The primitive non-associative rewrite rule: $(a \bullet b) \bullet c \rightarrow a \bullet (b \bullet c)$. This is the coupling signature for choice resolution, lifted through subtrees.

Definition 4 (Multi-Step Contraction). The reflexive-transitive closure of **contracts_one**. Represents an audit trail of choice resolutions with monotonic provenance.

Definition 5 (Right-Comb Normal Form). The minimum element of the Tamari lattice — the equilibrium attractor. Defined as **rightComb**(0) = **Leaf** and **rightComb**($n+1$) = **Node**(**Leaf**, **rightComb**(n)).

1.2 Lifting Lemmas

Lemma 6 (Contraction Preserved Under Left Node). *If $t \rightarrow t'$ then $\text{Node}(t, r) \rightarrow \text{Node}(t', r)$. Choice resolution in the left subtree propagates to the whole composition.*

Lemma 7 (Contraction Preserved Under Right Node). *If $r \rightarrow r'$ then $\text{Node}(l, r) \rightarrow \text{Node}(l, r')$. Choice resolution in the right subtree propagates to the whole composition.*

Lemma 8 (Transitivity of Contraction). *If $s \rightarrow t$ and $t \rightarrow u$ then $s \rightarrow u$. Audit trails concatenate monotonically.*

1.3 Size Invariants

Lemma 9 (Size is Preserved by Single-Step Contraction). *Rotation rewrites the tree depth but does not change the number of internal nodes.*

Lemma 10 (Size is Preserved by Multi-Step Contraction). *Contraction paths preserve tree size, establishing that the Tamari lattice T_n is graded by node count n .*

1.4 Main Contraction Theorem

Lemma 11 (Node of Right-Combs Contracts to Right-Comb). *For any $a, b : \mathbb{N}$,*

$$\text{Node}(\text{rightComb}(a), \text{rightComb}(b)) \longrightarrow \text{rightComb}(1 + a + b).$$

The equilibrium attractor is closed under non-associative composition.

Theorem 12 (Every Tree Contracts to Right-Comb). *For any tree $t : \text{EMLTree}$,*

$$t \longrightarrow \text{rightComb}(t.\text{size}).$$

Every configuration has a valid evolution path to the equilibrium attractor indexed by its size.

1.5 Router and Registry

Definition 13 (Router Index). A bounded natural number $\text{Fin}(n)$ serving as a neural binding address.

Definition 14 (Type Registry). An injective mapping from router indices to EML trees. This is the cortex-registry interface where neural computation meets formal grammar.

1.6 Witness-Skeptic Game

Definition 15 (Cortex Certificate). A proof-carrying audit trail $\langle \text{source}, \text{target}, \text{proof} \rangle$ showing that a tree reaches its equilibrium.

Definition 16 (Certify). Issues a `CortexCertificate` for any tree t , certifying that $t \rightarrow \text{rightComb}(t.\text{size})$. This is the quench witness in the Witness-Skeptic game.

Chapter 2

Logic Types: Pluralistic Reasoning Framework

2.1 Logic Type Hierarchy

Definition 17 (13 Logic Types). An inductive type enumerating 13 reasoning paradigms. Each handles specific classes of paradoxes as boundary conditions rather than errors.

Definition 18 (Logic Classification). Classification of logic types as Substructural, Extension, Alternative, Generalization, or Classical.

Definition 19 (Logic Tree). A type family assigning EML trees to each logic type. Currently uniform; extensible to logic-specific tree types.

Definition 20 (Logic Contraction). A type family of contraction relations indexed by `LogicType`. For Classical logic this is the Tamari contraction from `EMLRegistry`.

Definition 21 (Logic Translation). A structure specifying forward and backward maps between logic types with soundness and completeness proofs.

Definition 22 (Meta-Contraction). Cross-logic contraction allowing reasoning that spans multiple logic types via intra-logic contraction, inter-logic translation, and transitivity.

2.2 Cayley-Dickson Connection

Definition 23 (CD Step). Maps logic types to Cayley-Dickson construction steps.

Definition 24 (Split-Octonion Sectors). Divides logic types into associative and non-associative sectors of the split-octonion algebra.

2.3 Normal Forms and Contraction Theorem

Definition 25 (Logic Normal Form). The normal form for each logic type. For Classical logic this is `rightComb`.

Theorem 26 (Logic Contraction to Normal Form). *For any logic type lt and tree t , `LogicContraction lt t` (`LogicNormalForm lt t`). Currently proven for Classical logic; other cases are placeholders.*

Chapter 3

Problem Classes and the Very Big Box

3.1 Paradox Taxonomy

Definition 27 (Problem Class). An inductive type enumerating 12 paradox classes, each with a native logic type: self-reference (ManyValued), vagueness (Fuzzy), inconsistent definition (Paraconsistent), temporal decision (Temporal), deontic (Deontic), epistemic (Epistemic), quantum superposition (Quantum), constructive (Intuitionistic), relevance (Relevance), empty reference (Free), infinity (Infinitary), modality (Modal).

Definition 28 (Problem). A structure parameterized by a problem class, a name, a list of suitable logics, a tree family $\text{LogicType} \rightarrow \text{EMLTree}$ (the paradox as encoded in each logic), and a normal form family $\text{LogicType} \rightarrow \text{EMLTree}$ (the resolution target for each logic).

Definition 29 (Wrapped Problem). The WfCA collapse rule: pairs a Problem with a LogicType. Records the tree (problem as interpreted in this logic), target (normal form), cost (pentagonator distance Φ), and proof (contraction path). A problem's **superposition** of possible resolutions collapses to the definite outcome given by the logic type's contraction rule.

3.2 The Liar as a Problem Instance

Definition 30 (Liar Problem). The Liar Paradox stated as a Problem of class `selfReference`. Its tree is the size-3 symmetric tree $\text{Node}(\text{Node}(\text{Leaf}, \text{Leaf}), \text{Node}(\text{Leaf}, \text{Leaf}))$ for most logics, except ManyValued which uses a size-2 left-comb encoding the three-valued disjunction $\text{True} \vee \text{False} \vee \text{Undefined}$. All logics target `rightComb(3)` as the normal form.

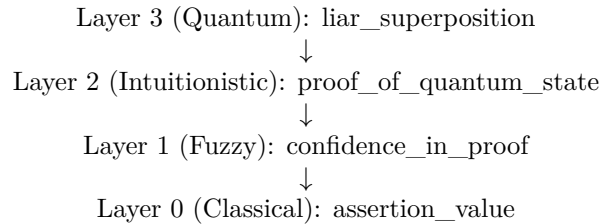
3.3 Prototype: Fuzzy Logic Wrapper

Definition 31 (Fuzzy Liar). The Liar wrapped in Fuzzy logic. The fixed point $\neg 0.5 = 0.5$ gives pentagonator distance $\Phi = 1$ (one collapse step). **Status:** The proof is a sorry placeholder because the Fuzzy [LogicContraction](#) is not yet defined. This is the WfCA prototype: a problem in superposition (all truth values $[0, 1]$) collapses to the unique fixed point under the Fuzzy contraction rule.

Definition 32 (Classical Liar). The Liar wrapped in Classical logic. Cost $\Phi = 0$ (at equilibrium). Proven using `contracts_to_rightComb`: the symmetric tree contracts to `rightComb(3)` in finitely many steps.

3.4 Nested Wrappers (Stacked Collapse)

Definition 33 (Tower). A sequence of logic-wrapped problems, each collapsing the previous layer’s output. Implements the nested wrapper pattern from Lumo section B:



Each layer’s logic type is tracked via a dependent pair Σlt , `WrappedProblem` p lt . The total cost is the sum of per-layer costs.

Definition 34 (Liar Tower). A four-layer prototype tower alternating Classical and Fuzzy wrappers. Demonstrates the stacked collapse pattern; the full tower with Quantum, Intuitionistic, Fuzzy, and Classical layers is not yet populated because non-Classical `LogicContractions` are missing.

3.5 The Very Big Box

The Very Big Box is the product space

$$\mathcal{V} = \prod_{c \in \text{ProblemClass}} \prod_{lt \in \text{suitableLogics}(c)} \text{WrappedProblem}(\text{problem}(c), lt)$$

where each coordinate is a problem in a specific logic and the Tower construction provides the nesting structure. Currently only the Liar ($c = \text{selfReference}$) is populated, and only its Classical wrapper is fully proven. The remaining coordinates represent the **real friction**: every cell in this product that lacks a proof is a concrete blocker.

3.6 Missing Work

The following definitions and theorems are stated but not yet implemented. They represent the **real friction** in the framework — the Liar Paradox is defined and its Classical resolution is proven, but cross-logic analysis cannot proceed until these pieces are populated.

3.6.1 Sorries in `LogicTypes.lean`

- **LogicContraction** for Fuzzy, ManyValued, Paraconsistent, Temporal, Deontic, Epistemic, Quantum, Intuitionistic, Relevance, Free, Infinitary, Modal — only Classical is defined (as the Tamari contraction from `EMLRegistry`).
- **LogicNormalForm** for all non-Classical logics — only `rightComb` (Classical) is defined.

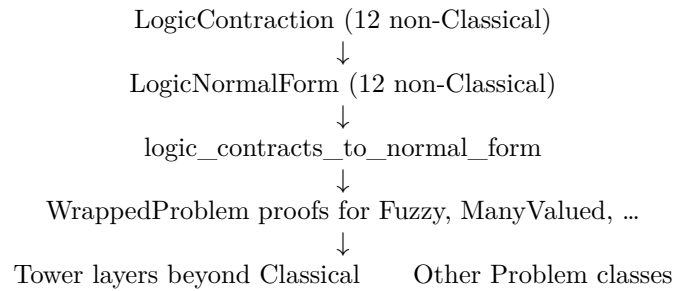
- **logic_contracts_to_normal_form** for non-Classical logics — the theorem is stated but only the Classical case is proven.

3.6.2 Stubs in `LiarParadox.lean`

- **fuzzyLiar.proof** — a sorry; blocked by the missing Fuzzy `LogicContraction`.
- **contractionCost** in the old version — the new `WrappedProblem` structure encodes cost directly; non-Classical costs are 0 because no logic-specific contraction exists to measure.
- **Non-Classical wrappers** for the Liar — `ManyValued`, `Paraconsistent`, `Intuitionistic`, etc. are not yet defined; each requires its logic’s `LogicContraction` and `LogicNormalForm`.
- **Other problem classes** — only `selfReference` (the Liar) is instantiated as a `Problem`. The other 11 classes have problems that are not yet defined.

3.6.3 Blocking Graph

The dependencies among missing work:



Every missing piece in the chain below `LogicContraction` is blocked by it. The Liar formalization makes this blocking structure explicit: previously the sorries in `LogicTypes.lean` were abstract “to be filled later” — now they are concrete dependencies of the Very Big Box product.